

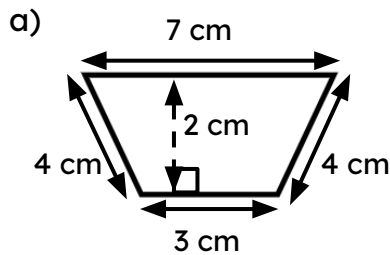


Calculating missing side lengths from the area of a trapezium

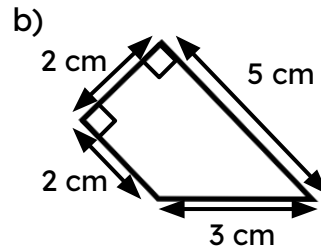
Task A: Calculating the area of a trapezium

$$\frac{(a + b) h}{2}$$

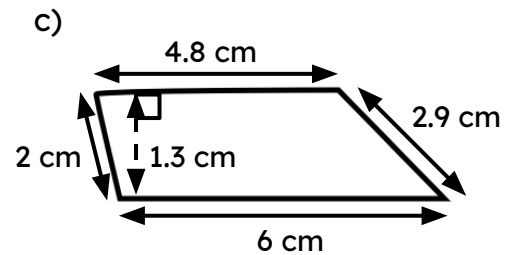
1) Calculate the area of each trapezium.



$$\begin{aligned} \text{Area} &= \frac{(3+7) \times 2}{2} \\ &= \frac{10 \times 2}{2} \\ &= \frac{20}{2} \\ &= 10 \text{ units}^2 \end{aligned}$$



$$\begin{aligned} \text{Area} &= \frac{(2+3) \times 2}{2} \\ &= \frac{7 \times 2}{2} \\ &= \frac{14}{2} \\ &= 7 \text{ units}^2 \end{aligned}$$



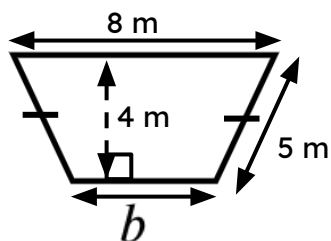
$$\begin{aligned} \text{Area} &= \frac{(4.8+6) \times 1.3}{2} \\ &= \frac{10.8 \times 1.3}{2} \\ &= \frac{14.04}{2} \\ &= 7.02 \text{ units}^2 \end{aligned}$$



Task B: Finding a missing side length using the area

1) Calculate the length of the marked side and then, if possible, calculate the perimeter.

a)



$$\text{Area} = 22 \text{ m}^2$$

$$22 = \frac{(b+8) \times 4}{2}$$

$$44 = (b + 8) \times 4$$

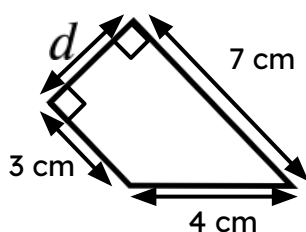
$$11 = (b + 8)$$

$$3 = b$$

The length is 3 m.

The perimeter is $8 + 5 + 5 + '3' = 21 \text{ m}$

b)



$$\text{Area} = 10 \text{ cm}^2$$

$$10 = \frac{(3+7) \times d}{2}$$

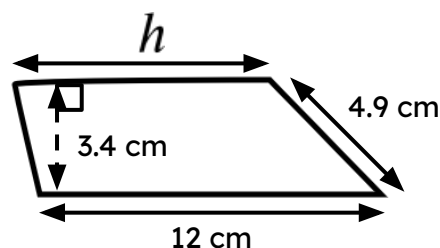
$$20 = 10 \times d$$

$$2 = d$$

The length is 2 cm.

The perimeter is $7 + 4 + 3 + '2' = 16 \text{ cm}$

c)



$$\text{Area} = 32.64 \text{ cm}^2$$

$$32.64 = \frac{(h+12) \times 3.4}{2}$$

$$65.28 = (h + 12) \times 3.4$$

$$19.2 = h + 12$$

$$7.2 = h$$

The length is 7.2 cm.

The perimeter cannot be found as one side length is still unknown.

